Roadmap. §41 (pp. 1–3): LSZ for b, d by LecNote4's argument; p. 3 normalizes Ψ. §42 (pp. 3–4): $S = i\langle 0|\mathcal{T}\Psi\bar{\Psi}|0\rangle$, the inverse Dirac operator of LecNote17's path integral, now from spin sums. §45 (pp. 4–7): $e^-\varphi \to e^-\varphi$ at tree level; $e^+\varphi$ begun. Next, LecNote20: charge-flow arrow, other Yukawa trees, spin sums (§§45–48). Supplements: AsymptoticStates, FeynmanPropagator, FeynmanRulesLSZ, FermionSigns, DiracFeynmanRules.

Lec19

"wave-packet" $b^\dagger$ and $d^\dagger$

$$b_1^\dagger \equiv \int d^3\vec{p}\, f_1(\vec{p})\, b_{s_1}^\dagger(\vec{p})$$

$$|i\rangle = \lim_{t\to-\infty} b_1^\dagger(t)\, b_2^\dagger(t)|0\rangle$$

$$|f\rangle = \lim_{t\to+\infty} b_1^\dagger(t)\, b_2^\dagger(t)|0\rangle$$

$\langle i|i\rangle = 1$, $\langle f|f\rangle = 1$.

for $\langle f|i\rangle$, we note (again, same as in scalar theory)

$$b_1^\dagger(-\infty) - b_1^\dagger(+\infty) = -\int_{-\infty}^{+\infty} dt\, \partial_0 b_1^\dagger(t)$$

$$= -\int d^3p\, f_1(\vec{p}) \int d^4x\, \partial_0\left(e^{-ipx}\bar{\psi}^+(x)\gamma^0 u_{s_1}(\vec{p})\right)$$

$$= -\int d^3p\, f_1(\vec{p}) \int d^4x\, \bar{\psi}(x)\left(\gamma^0\overleftarrow{\partial_0} - i\gamma^0 p^0\right) u_s(\vec{p})\, e^{-ipx}$$

$$\gamma^0\overleftarrow{\partial_0} - i\gamma^i p^i - im$$

$$\gamma^0\overleftarrow{\partial_0} - \gamma^i\overleftarrow{\partial_i} - im$$

(IBP).

$$\gamma^0\overleftarrow{\partial_0} + \gamma^i\overleftarrow{\partial_i} - im$$

$$= i\int d^3p\, f_1(\vec{p}) \int d^4x\, \bar{\psi}(x)\left(+i\overleftarrow{\not{\partial}} + m\right) \cancel{u_s(\vec{p})}\, u_s(\vec{p})\, e^{-ipx}$$

Recall in scalar case

$$a_1^\dagger(-\infty) - a_1^\dagger(+\infty) = -i\int d^3k\, f_1(\vec{k}) \int d^4x\, e^{ikx}(-\partial^2 + m^2)\varphi(x)$$ [1]

[1] Should read $a_1^\dagger(-\infty) - a_1^\dagger(+\infty) = +i\int d^3k\, f_1 \int d^4x\, e^{ikx}(-\partial^2 + m^2)\varphi$: the same sign as the fermion line above and as $a^\dagger(\mathbf{k}) \to i\int\cdots$ on p. 5 (LecNote4, Srednicki §5).

In free field theory $\bar{\Psi}(+i\overleftarrow{\not\partial}+m)=0$
so the particle propagation is only changed when interaction is on.

Similarly

$$b_1(+\infty)-b_1(-\infty)=-i\int d^3p\, f_1(\vec{p})\int d^4x\, e^{-ipx}\bar{u}_{s_1}(\vec{p})(-i\not\partial+m)\Psi(x)$$

$$d_1^\dagger(+\infty)-d_1^\dagger(-\infty)=-i\int d^3p\, f_1(\vec{p})\int d^4x\, e^{ipx}\bar{v}_{s_1}(\vec{p})(-i\not\partial+m)\Psi(x)$$ ①

$$d_1(+\infty)-d_1(-\infty)=-i\int d^3p\, f_1(\vec{p})\int d^4x\,\bar{\Psi}(x)(+i\overleftarrow{\not\partial}+m)v_1(\vec{p})\,e^{-ipx}$$

1 Should read $\bar{v}_{s_1}(\mathbf{p})(-i\not\partial+m)\Psi(x)$: the overwritten sign is $-$, as on p. 7, since $\bar{v}(-\not p+m)=0$. The left side is $d_1^\dagger(-\infty)-d_1^\dagger(+\infty)$.

Now (for $2\to2$ of $+$ type)

$$\langle f|i\rangle=\langle 0|b_{2'}(+\infty)b_{1'}(+\infty)b_1^\dagger(-\infty)b_2^\dagger(-\infty)|0\rangle$$

$$=\langle 0|\overline{T}\,b_{2'}(+\infty)b_{1'}(+\infty)b_1^\dagger(-\infty)b_2^\dagger(-\infty)|0\rangle$$

$\overline{T}$: here change order gives extra minus sign for fermions for odd # of changes.

$$=i^{-4}\int d^4x_1\,d^4x_2\,d^4x_{1'}\,d^4x_{2'}$$

$$e^{-ip_{1'}x_{1'}}\,[\bar{u}_{s_{1'}}(\vec{p}_{1'})(-i\not\partial_{1'}+m)]_{\alpha_1}$$

$$e^{-ip_{2'}x_{2'}}\,[\bar{u}_{s_{2'}}(\vec{p}_{2'})(-i\not\partial_{2'}+m)]_{\alpha_2}$$

$$\langle 0|\overline{T}\,\bar{\Psi}_{\alpha_2}(x_{2'})\bar{\Psi}_{\alpha_1}(x_{1'})\bar{\Psi}_{\alpha_1}(x_1)\bar{\Psi}_{\alpha_2}(x_2)|0\rangle$$

$$[(+i\overleftarrow{\not\partial}_1+m)u_{s_1}(\vec{p}_1)]_{\alpha_1}e^{-ip_1x_1}$$

$$[(+i\overleftarrow{\not\partial}_2+m)u_{s_2}(\vec{p}_2)]_{\alpha_2}e^{-ip_2x_2}$$

(3) (16)

1 Of the lines with Ψ, only the v and $\bar{u}$ ones are conditions. $\langle 0|\Psi|0\rangle = \langle 0|\bar{\Psi}|0\rangle = 0$ are automatic (a 2π rotation flips Ψ), as are $\langle p,s,+|\Psi|0\rangle = \langle p,s,-|\bar{\Psi}|0\rangle = 0$ (Ψ lowers the charge by one). The two fix the normalization of Ψ and make the m in u, v and LSZ the physical mass (AsymptoticStates).

Similar to scalar case, consistency requires that (1)

$\langle 0|0\rangle = 1$

$\langle 0|\Psi(x)|0\rangle = 0, \quad \langle 0|\bar{\Psi}(x)|0\rangle = 0$

$\langle p,s,+|\Psi(x)|0\rangle = 0$

$\langle p,s,-|\Psi(x)|0\rangle = v_s(\vec{p})\, e^{-ipx}$

$\langle p,s,+|\bar{\Psi}(x)|0\rangle = \bar{u}_s(\vec{p})\, e^{ipx}$ (2)

$\langle p,s,-|\bar{\Psi}(x)|0\rangle = 0$

Recall $\Psi(x) = \ldots b_s(\vec{p}) \ldots d_s^\dagger(\vec{p}) \ldots$

2 Should read $\bar{u}_s(\mathbf{p})\, e^{-ipx}$: $\bar{\Psi} \ni b^\dagger \bar{u}\, e^{-ipx}$. Translation invariance gives e^{-ipx} for every $\langle p|\mathcal{O}(x)|0\rangle$, as in the v line above.

One can further see that

$$S(x-y)_{\alpha\beta} \equiv i\langle 0|T\Psi_\alpha(x)\bar{\Psi}_\beta(y)|0\rangle$$

$$= \not{A} \int \frac{d^4p}{(2\pi)^4} e^{-ip(x-y)} \frac{(-\not{p}+m)_{\alpha\beta}}{p^2+m^2-i\varepsilon}$$

And $(-i\not{\partial}_x + m)_{\alpha\beta} S(x-y)_{\beta\gamma} =$

$$\int \frac{d^4p}{(2\pi)^4} e^{-ip(x-y)} \frac{(\not{p}+m)_{\alpha\beta}(-\not{p}+m)_{\beta\gamma}}{p^2+m^2-i\varepsilon}$$

$$= \int \frac{d^4p}{(2\pi)^4} e^{-ip(x-y)} \frac{(p^2+m^2)\,\delta_{\alpha\gamma}}{p^2+m^2-i\varepsilon}$$

$$= \delta^4(x-y)\,\delta_{\alpha\gamma}.$$

$$S(x-y)_{\alpha\beta}(+i\overleftarrow{\not{\partial}}_y + m)_{\beta\gamma} = \delta^4(x-y)\,\delta_{\alpha\gamma}.$$

To demonstrate.

$$\langle 0|\bar{\Psi}_\alpha \bar{\Psi}_\beta|0\rangle = \sum_{s,s'} \int \widetilde{dp}\, \widetilde{dp'}\, e^{-ipx} e^{-ip'y} u_{s\alpha}(\vec{p})\, \bar{u}_{s'\beta}(\vec{p}')\, \langle 0|b_s(\vec{p})\, b_{s'}^\dagger(\vec{p}')|0\rangle$$

$$= \sum_{s,s'} \int \widetilde{dp}\, \widetilde{dp'}\, e^{-ip(x-y)} u_{s\alpha}(\vec{p})\, u_{s'\beta}^*(\vec{p}') \qquad \downarrow \quad (2\pi)^3 2\omega\, \delta_{ss'}$$

$$= \int \widetilde{dp}\, e^{-ip(x-y)} (-\not{p}+m)_{\alpha\beta}$$

and also

$$\langle 0|\bar{\Psi}_\beta \bar{\Psi}_\alpha|0\rangle = \cdots \langle 0|d_{s'}(\vec{p}')\, d_s^\dagger(\vec{p})|0\rangle$$

$$= \int \widetilde{dp}\, e^{-ip(x-y)} (-\not{p}-m)_{\alpha\beta}$$ (1)

Now Finally we have everything:

Taking the example of $e^- \varphi \to e^- \varphi$.
(b-type) (b-type)

$$\langle f|i\rangle = \langle 0|T\, a_{out}(\vec{k}')\, b_{s',out}(\vec{p}')\, b_{s,in}^\dagger(\vec{p})\, a_{in}^\dagger(\vec{k})|0\rangle.$$

plug-in $b_s^\dagger(\vec{p})_{in} \to i\int d^4y\, \bar{\Psi}(y)(+i\overleftarrow{\not{\partial}}+m) u_s(\vec{p}) e^{ipy}$

$b_{s'}(\vec{p}')_{out} \to i\int d^4x\, e^{-ip'x}\, \bar{u}_{s'}(\vec{p})(-i\not{\partial}+m)\Psi(x)$

1 Why these give p. 3's S:
$\mathcal{T}\Psi(x)\bar{\Psi}(y) \equiv \theta(x^0-y^0)\Psi\bar{\Psi} - \theta(y^0-x^0)\bar{\Psi}\Psi$. The fermionic minus cancels the one in $-\not{p}-m$: both terms are $(i\,\partial\!\!\!/_x + m)$ on the scalar integrals, so $i\langle\mathcal{T}\Psi\bar{\Psi}\rangle = (i\,\partial\!\!\!/_x + m)\Delta = S$, $-i\epsilon$ included. The jump at $x^0 = y^0$ is $\{\Psi, \bar{\Psi}\} = \gamma^0\delta^3$ (LecNote18 p. 10); with $+$, no Green's function.

$$a^+(\vec{k}) \to i\int d^4z_1\, e^{-ikz_1}(-\partial^2+m^2)\varphi(z_1)$$

$$a(\vec{k}') \to i\int d^4z_2\, e^{+ik'z_2}(-\partial^2+m^2)\varphi(z_2)$$

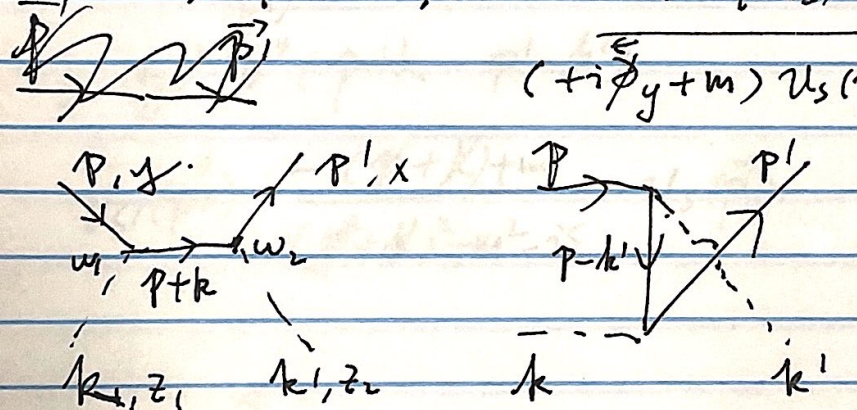
We have

$$\langle f|i\rangle = (i)^4\int d^4x\, d^4y\, d^4z_1\, d^4z_2\, e^{i(py-px+kz_1-k'z_2)}$$

$$(-\partial_{z_1}^2+m_\varphi^2)(-\partial_{z_2}^2+m_\varphi^2)$$

$$\bar{u}_{s'}(\vec{p}')(-i\not{\partial}_x+m)\langle 0|T\,\varphi(z_2)\bar{\psi}(x)\bar{\psi}(y)\varphi(z_1)|0\rangle$$

$$(+i\overleftarrow{\not{\partial}}_y+m)u_s(\vec{p})$$



$$\int d^4w_1\, d^4w_2$$

$$i\,i^2\left(\frac{1}{i}\right)^4(ig)^2\, S(x-w_2)\,\Delta(z_2-w_2)\,S(w_2-w_1)\,\Delta(z_1-w_1)\,S(w_1-y)$$ [1]

$S(x-w_2)$ → $(-i\not{\partial}_x+m)$

$\delta(z_2-w_2)$ → $(-\partial_{z_2}^2+m_\varphi^2)$

$S(w_2-w_1)$

$\delta(z_1-w_1)$

$\delta(w_1-y)$

Then $\int d^4x\, d^4y\, d^4z_1\, d^4z_2\, e^{-i\cdots}$

→

1 The prefactor is $i^4(\frac{1}{i})^5(ig)^2 = \frac{1}{i}(ig)^2$: i^4 from LSZ, $\frac{1}{i}$ per propagator ($\langle\mathcal{T}\Psi\bar{\Psi}\rangle = S/i$, $\langle\mathcal{T}\varphi\varphi\rangle = \Delta/i$), ig per vertex; the $\frac{1}{2!}$ of $e^{i\int\mathcal{L}_1}$ cancels against $w_1 \leftrightarrow w_2$ (LecNote18 p. 1's correlator). No t-channel: each vertex has one φ, so the two φ's sit on different vertices.

Knowing $S(\omega_2-\omega_1) = \int \frac{d^4p''}{(2\pi)^4} e^{-ip''(\omega_2-\omega_1)}$

$$\frac{(-\not{p}''+m)}{p''^2+m^2-i\varepsilon}$$

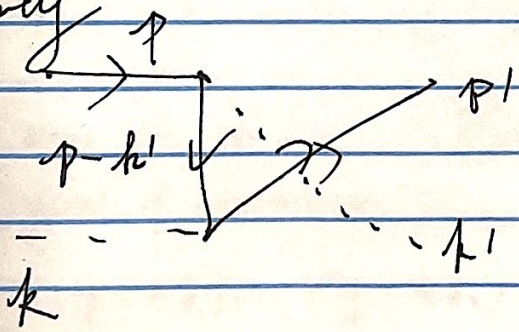
$$\rightarrow \langle f|i\rangle \ni \frac{1}{i}(ig)^2 \int d^4\omega_1 d^4\omega_2 \; \bar{u}_{s'}(\vec{p}') \int \frac{d^4p''}{(2\pi)^4} e^{-ip''(\omega_2-\omega_1)}$$

$$e^{i(p+k)\omega_1 - (p'+k')\omega_2} \; \bar{u}_{s'}(\vec{p}') \frac{-\not{p}''+m}{p''^2+m^2-i\varepsilon} u_s(\vec{p})$$

$$= \frac{1}{i}(ig)^2 \cdot (2\pi)^4 \delta^4(p+k-p'-k')$$

$$\bar{u}_{s'}(\vec{p}') \frac{-(\not{p}+\not{k})+m}{(p+k)^2+m^2-i\varepsilon} u_s(\vec{p})$$

Similarly



has

$$\frac{1}{i}(ig)^2 \; \bar{u}_{s'}(\vec{p}') \frac{-(\not{p}-\not{k}')+m}{(p-k)^2+m^2-i\varepsilon} u_s(\vec{p}).$$ (1)

1 Should read $(p-k')^2+m^2-i\epsilon$ (no slash; k' as in the numerator). With $\langle f|i\rangle = (2\pi)^4\delta^4\, i\mathcal{T}$: $i\mathcal{T} = \frac{1}{i}(ig)^2\bar{u}'[\tilde{S}(p+k)+\tilde{S}(p-k')]u$, $(p+k)^2 = -s$, $(p-k')^2 = -u$.

[1] Rules (§45): vertex ig; internal line $\frac{1}{i}\tilde{S}(q)$, q along the arrow (e^+ momentum runs against it, below); external u (e^- in), $\bar{u}$ (e^- out), $\bar{v}$ (e^+ in), v (e^+ out), read against the arrow; -1 per closed fermion loop; relative signs from fermion exchange, overall sign unphysical. LecNote20 p. 1 derives the arrow.

We can see the "Rules" here [1]

But it is slightly more subtle than that as (there are ϕ-type)

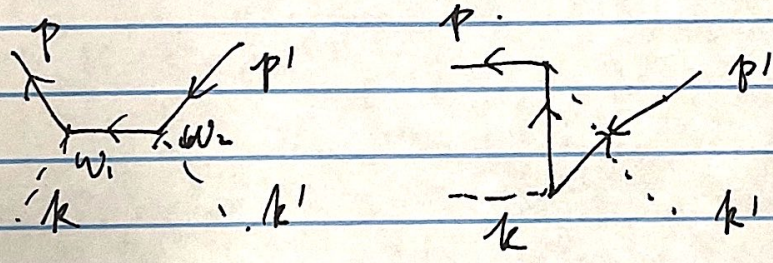
For $e^+\varphi \to e^+\varphi$.

$$\langle f|i\rangle = \langle 0|\, T\, a(\vec{k}')\, d_{s'}(\vec{p}')\, d_s^\dagger(\vec{p})\, a^\dagger(\vec{k})|0\rangle.$$

$$d_s^\dagger(\vec{p})_{in} \to -i\int d^4x\, e^{+ipx}\, \bar{v}_s(\vec{p})(-i\not{\partial}+m)\bar{\psi}(x)$$

$$d_s(\vec{p}')_{out} \to -i\int d^4y\, \bar{\psi}(y)(+i\overleftarrow{\not{\partial}}+m)v_{s'}(\vec{p}')\, e^{-ip'y}$$

$$\vdots$$



the "sign in p" is flipped and hence the "flow of momentum"

$$e^{i(px+kz_1-p'y-k'z_2)}$$

$$\downarrow$$

$$e^{i(p+k)w_1 - i(p'+k')w_2}\, e^{ip''(w_2-w_1)}$$ [2]

[2] Finishing: the arrow runs $w_2 \to w_1$, so the line is $S(w_1-w_2) = \int e^{ip''(w_2-w_1)}\tilde{S}(-p'')$, $p'' = p+k$:

$$i\mathcal{T} = \tfrac{1}{i}(ig)^2\,\bar{v}_p\Big[\frac{\not{p}+\not{k}+m}{-s+m^2} + \frac{\not{p}-\not{k}'+m}{-u+m^2}\Big]v_{p'},$$

up to an overall sign ($\bar{\Psi}(y)\Psi(x) = -\Psi(x)\bar{\Psi}(y)$ in $\mathcal{T}$). Checked: same spin-summed $|\mathcal{T}|^2$ as $e^-\varphi$ (C invariance).